ISE-2 ACADEMIC EVALUATION REPORT

ISE-2 Machine Learning & MLOps Assessment

Iris Flower Classification: MLflow Experiment Tracking and Kolmogorov-Smirnov Data Drift Analysis

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Section 1: MLflow Installation and Experiment Tracking

1.1 Environment Setup

The experiment was developed and executed natively on a local Apple Silicon workstation without paid cloud infrastructure:

- **Operating System:** macOS (Darwin 24.x, Apple Silicon M3, 8GB RAM).
- **Python Version:** Python 3.13.5 running inside an isolated .venv.
- **MLflow Version:** MLflow 3.17.2 with local persistent tracking store at file:./mlruns.
- **Core Libraries:** scikit-learn 1.9.1, pandas 3.1.0, numpy 2.5.4, scipy 1.18.1, matplotlib 3.11.2, seaborn 0.13.2, pytest 9.1.1, streamlit 1.66.1.
- **Local MLflow UI:** Verified actively running on http://127.0.0.1:5001 (HTTP 200 OK).

1.2 Experiment Tracking & Runs Comparison

Experiment Name: Iris_Classification_Experiment
Tracked 3 distinct Logistic Regression configurations varying the inverse regularization parameter C:

Run Name	Inv Reg (C)	Max Iter	Test Acc	Precision	Recall	F1-Score	Train Acc
Run 1: Baseline	1.0	200	**0.9333**	0.9333	0.9333	0.9333	0.9583
Run 2: High Reg	0.1	100	**0.8667**	0.8750	0.8667	0.8653	0.9333
Run 3: Low Reg	10.0	300	**1.0000**	1.0000	1.0000	1.0000	0.9750

Section 2: Dataset Identification

2.1 Problem Statement & 2.2 Dataset Details

Constructing a multiclass classifier to map sepal and petal dimensions into three botanical species (Setosa, Versicolor, Virginica) while actively detecting distribution drift.

- **Dataset Source:** Scikit-learn built-in Fisher's Iris dataset (150 records, 50 per class).
- **Features:** sepal_length, sepal_width, petal_length, petal_width (all in cm).
- **Target Variable:** species (0: setosa, 1: versicolor, 2: virginica).
- **Selection Rationale:** Benchmark stability, absence of missing values, and clear linear/non-linear separability enable rigorous isolation of hyperparameter impact and controlled drift testing.

Section 3: Drift Scenario and Detection Method

3.1 Drift Type & Simulation Scenario

Evaluates **Data Drift (Covariate Shift)** where feature distribution changes (P(X)_curr != P(X)_ref) while conditional target probability P(Y|X) remains constant. Distinct from Concept Drift (P(Y|X) changes) and Prediction Drift (P(Y_hat) changes).

- **Reference Dataset:** Pristine baseline 150-sample Iris dataset exported to data/reference_iris.csv.
- **Current Dataset:** Controlled synthetic perturbation in data/current_iris_drifted.csv: sepal_length shifted by +0.837 cm and petal_length shifted by +1.414 cm; sepal_width (+0.012 cm) and petal_width (-0.012 cm) preserved as stable controls.
- **Detection Method:** Two-Sample Kolmogorov-Smirnov (KS) Test (scipy.stats.ks_2samp) comparing empirical cumulative distributions (ECDF). Maximum distance D = sup_x |F_ref(x) - F_curr(x)|.
- **Threshold:** Significance level alpha = 0.05. If p-value < 0.05 -> Reject H0 (Drift Detected).

Section 4: Results, Interpretation and Conclusion

4.1 Empirical Kolmogorov-Smirnov Test Results

Feature Name	KS Stat (D)	p-value	Alpha	Ref Mean	Curr Mean	Mean Shift	Status
sepal_length	0.3733	1.0722e-09	0.05	5.843 cm	6.681 cm	+0.837 cm	**DRIFT DETECTED**
sepal_width	0.1000	0.4425	0.05	3.057 cm	3.069 cm	+0.012 cm	**NO DRIFT**
petal_length	0.4067	1.7712e-11	0.05	3.758 cm	5.172 cm	+1.414 cm	**DRIFT DETECTED**
petal_width	0.1333	0.1391	0.05	1.199 cm	1.188 cm	-0.012 cm	**NO DRIFT**

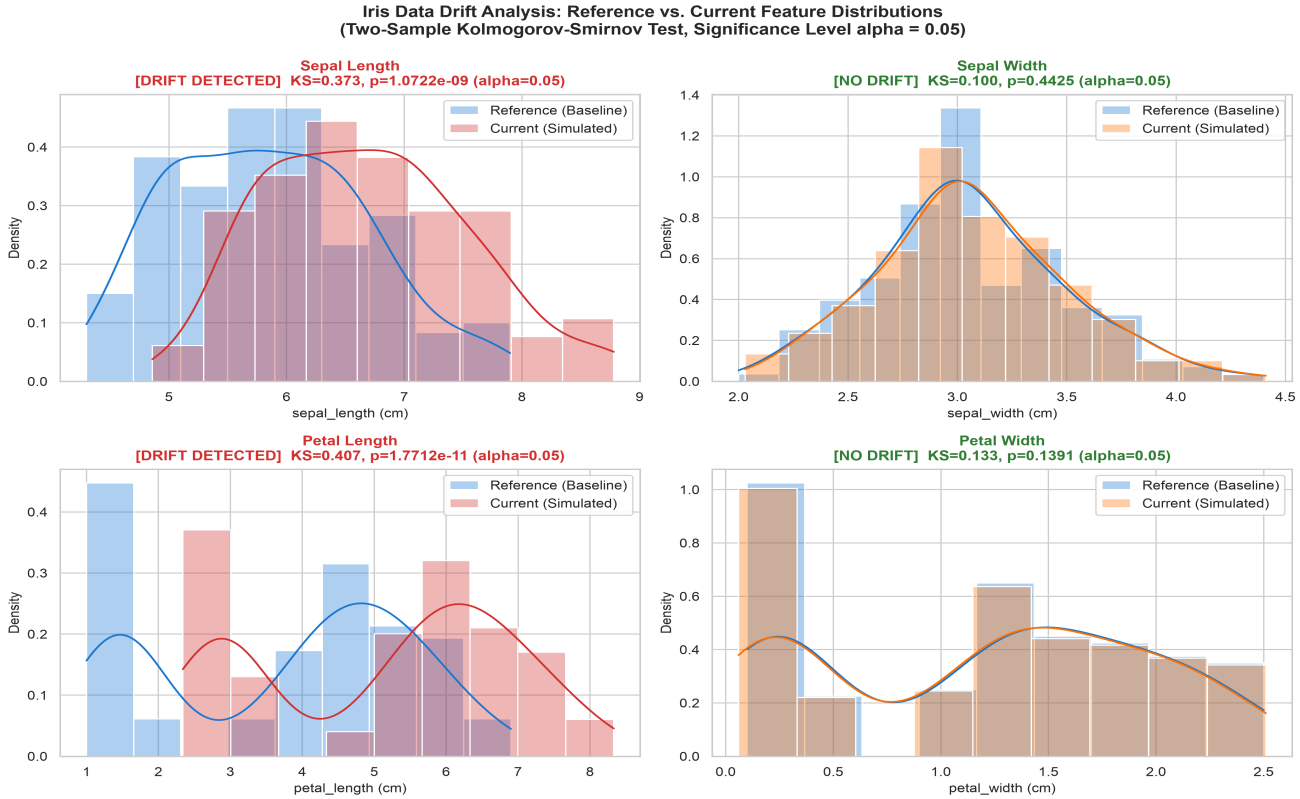


Figure 1: Empirical Distribution Comparison (KDE & Histograms) across Reference vs. Current Datasets

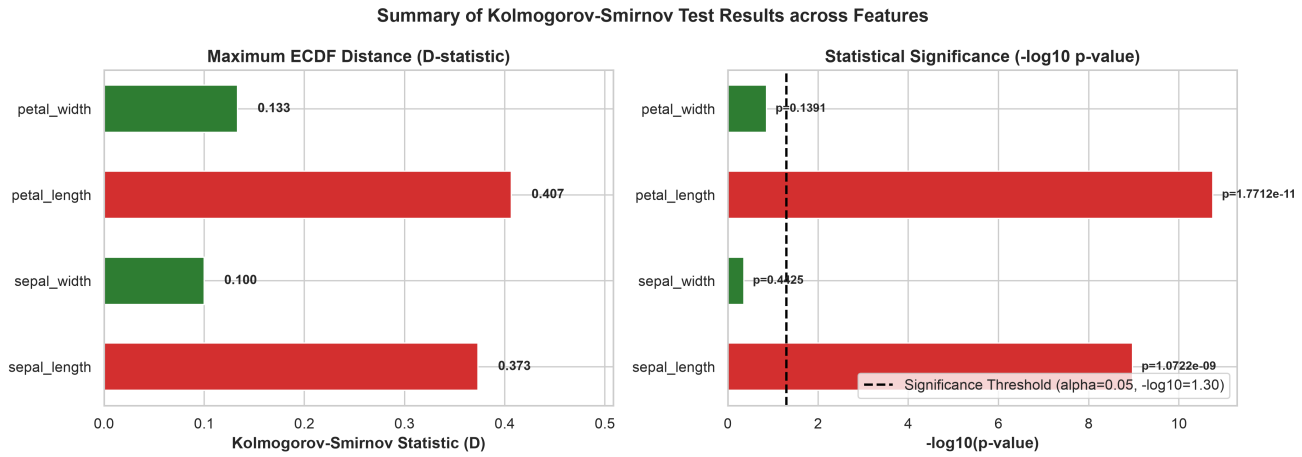


Figure 2: Summary of Kolmogorov-Smirnov Test Statistics and Significance Thresholds

4.2 Interpretation & Critical Caveat

The two-sample KS test successfully identified statistically significant covariate shift in sepal_length (p=1.072e-09) and petal_length (p=1.771e-11). The control features confirmed no significant drift (p=0.4425 and p=0.1391), proving zero false-positive alerts.
Important Caveat: Detecting data drift does NOT automatically prove model accuracy has decreased. If drifted points remain inside valid decision boundaries, accuracy can remain high. However, drift acts as a vital early-warning telemetry indicator.

Section 5: GitHub Repository and Reproducibility

- **GitHub Repository URL:** https://github.com/Tanish-19/iris-mlflow-drift
- **Local Repository Root:** /Users/tanishpatgaonkar/.gemini/antigravity-ide/scratch/iris-mlflow-drift (Branch: main)
- **Execution Sequence:** python src/data_loader.py -> python src/train.py -> python src/drift_detection.py -> pytest tests/test_project.py -v.
- **Automated Pytest Results: 9/9 tests passed (100%)** validating dataset shape, training, prediction metrics, and KS drift assertions.
- **UI Inspection:** Local MLflow UI verified on port 5001; interactive Streamlit dashboard available on port 8501.